

Cross-Platform Entertainment Recommendation System Documentation

Prerequisites:

Required Accounts:

- Google Account: For Google Sheets integration and Colab access
- Reddit Developer Account: For API Access (credentials are provided in script)
- Twitch Developer Account: For game popularity data collection
- TMDB Account: For movie metadata access
- Jikan API (Anime): no account required, free access

Required Files:

- *No files need to be uploaded*
- Reddit API Credentials: Already configured in the script
- Required CSV files will automatically be saved

API Credentials (Pre-Configured):

Reddit API:

- CLIENT_ID = "HA15MkoZrg4xng2Y7umGsA"
- CLIENT_SECRET = "yOJTSrreZG8rruSLOEOxi4XkASS7bw"
- USER_AGENT = 'game_data_collector_v1.0'

Twitch API:

- CLIENT_ID = 'b45lzn61sha4snodwzfzpl2e9supg7w'
- CLIENT_SECRET = 'ivoga8yqvcdrdxkop1jkq7eobe5ds5'

TMDB API:

- API_KEY = 'c1125eec3c170f3e54968a9c9dd5626c'

Google Sheets:

- SHEET_ID = '1wu53RWdFsxgq6DfglcPTmkxI7QPSw8mTjjAm_1oFmQk'

Data Sources & Database:

Primary Sources:

- MyAnimeList (via JikanPy): ANime ratings, genre, demographics, user lists
- Twitch API: Game libraries, playtime, reviews, genre classifications
- TMDB API: Movie ratings, genres, cast, user watchlists
- Reddit API (PRAW): Cross-Platform discussion data

Target Subreddits for Cross-Platform Analysis:

- General: r/anime, r/gaming, r/movies

- Gaming subgenres: r/JRPG, r/visualnovels
- Recommendation-focused: r/MovieSuggestions, r/gamingsuggestions
- Cross-reference: r/tipofmytongue

Data Collection Strategy:

- Scrape posts mentioning multiple entertainment types
- Extract consumption patterns from recommendation requests and discussions
- Collect metadata on content: genres, themes, art style, narrative elements
- Focused on the last 3 months of data for recent consumption trends
- Data processed in Google Sheets with automatic backup

Algorithm Development:

Part 1: Content-Based Filtering

Create content DNA profiles using:

- Genre Classifications (sci-fi, fantasy, romance)
- Narrative themes (coming-of-age, revenge, friendship)
- Art/Visual style indicators
- Target demographic alignment
- Pacing and tone characters

Part 2: Collaborative Filtering

- User-based collaboration filtering using consumption overlap patterns
- Matrix factorization for sparse data handling
- Implicit feedback from discussion engagement and mention frequency

Part 3: Hybrid Approach

- Combine content-based and collaborative filtering
- Weight recommendations based on confidence scores
- Boost trending content across platforms

Content DNA Creation:

- Automated Feature Extraction
 - NLP with spaCy: Extract thematic elements from descriptions and reviews
 - Keyword Clustering: For genre and theme identification
 - Mood Profiles: Based on discussions and reviews
- Cross-Platform Mapping
 - Genre translation dictionary
 - Theme correlation matrices between media
 - Demographic overlap and preference mapping

Recommendation System:

- Input Methods
 - Single Item: “Attack On Titan”
 - Multi-item profile: User provides favorite across platforms
 - Specifications: “Similar Movie”
- Output Format
 - Number of recommendation you want with confidence scores
 - Explanations for recommendations
 - Cross-Platform pathway suggestions

Setting up Google Colab:

Step 1: Access the Notebook

1. Navigate to project folder in Google Drive
2. Open the Colab notebook
(Cooledtured_CrossPlatformEntertainment_RecommendationSystem.ipynb)
3. Click “CONNECT” in the top right corner
4. Wait for connection - Green checkmark indicates ready

Step 2: Set Runtime Type

1. Click “Runtime” to “Change runtime type”
2. Select hardware accelerator:
 - a. CPU: Sufficient for this project
 - b. GPU: Optional
3. Click “Save” - Session will restart if changes are made

What Actually Happens When You Run:

Google Authentication:

1. Display a link in the output
2. You click the link and sign in with Google
3. Copy the authorization code
4. Paste it back into Colab
5. Done!

Data Flow:

- Part 1: Uses this Google authorization to write TO Google Sheets
- Part 2: Can either read FROM Google Sheets (using same authorization)
OR process the CSV files directly

Common Setup Issues:

Authentication & API Issues:

- Script fails to run/ stops at authentication step? (Invalid credentials or 401 Unauthorized)
 - Verify API credentials in the script cells
 - API tokens may be expired or revoked
 - Ensure no typos in credential strings

Data Collection/Processing Issues:

- Low Data Collection/ Empty Datasets:
 - Verify internet connection is stable
 - Check if API limits are reached
 - Ensure subreddit names are spelled correctly
- Errors in Content DNA creation/ NLP script fails to detect themes or genres:
 - Ensure all required Python packages are installed
 - Re-run data collection to regenerate CSV files

Dashboard Issues:

- Gradio fails to load/ Visualizations missing or incomplete:
 - Ensure all CSV files exist and were generated successfully
 - Reinstall/Update Gradio
 - Check logs from prior cells for errors

Performance Expectations:

Processing Time:

- Package Installation: 3-5 minutes
- Data Scraping per Subreddit/API: 15-20 minutes
- Content DNA & Recommendation Processing: 5-10 minutes
- Dashboard Generation: 2-3 minutes
- Total Runtime: 30-40 minutes

Expected Output:

- Multiple CSV files per platform/subreddit
- Cross-platform consumption datasets
- Interactive Dashboard with visualizations
- Recommendation lists with explanations

Dashboard Features:

- Interactive dashboard: Gradio
- Filters: Subreddit, date range, platform
- Search: By anime, game, or movie title
- Visualizations: Popularity rankings, sentiment/mood trends, cross-platform maps
- Exports: Download filtered datasets as CSV